

SUMMER INTERNSHIP ASSIGNMENTS

Assignment-1:-

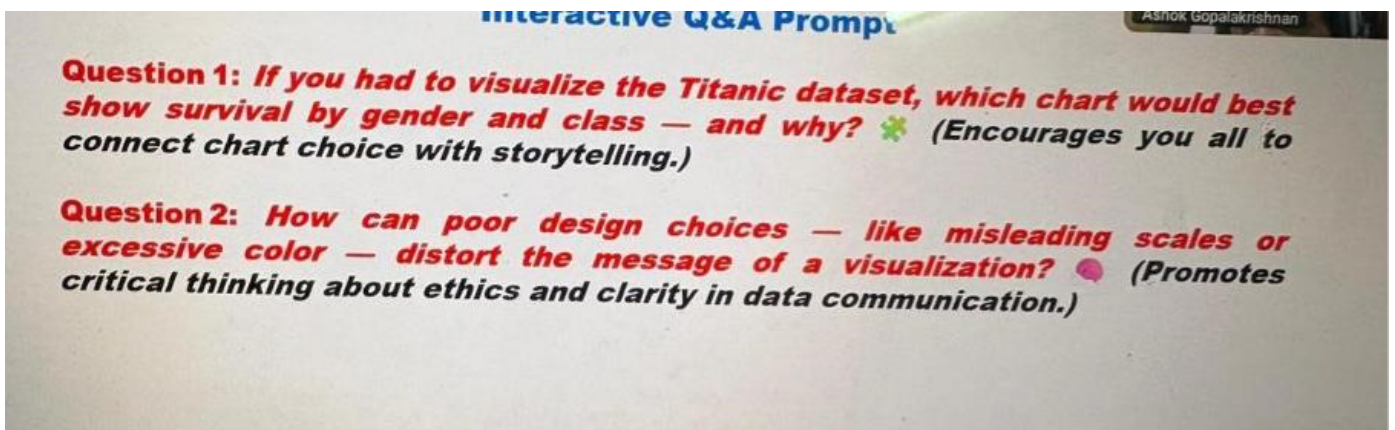
15/june

Download anaconda jupyter notebook and run simple code

Assignment-2:-

17/june

1. Answer the following Questions.



Titanic dataset link :- <https://share.google/fFY2IyxIK8iBqUVA7>

2. Write the code for a Library Management System and provide a summary of its logic

Assignment-3:-

18/June

Answer the following Questions.(Part 1-3)

Part-1

Statistics for Machine Learning

Questions for today, the 18 June 2026

- **What have you learnt from the Titanic Disaster?**
- **Distractions are dangerous, elaborate**
- **Stakeholders should be kept informed, yes/no**
- **Traceability is essential, what do you think?**
- **Documentation may have lasting benefits, true or false and why**

Our investigation aims to answer the following questions:

- 1. What were the key factors that influenced passenger survival on the Titanic?***
- 2. How did variables such as age, gender, ticket class, and family size affect the chances of survival?***
- 3. Can we identify significant predictors of survival based on the available data?***
- 4. Are the differences in survival rates of different demographics really statistically significant to be able to make conclusions from it?***

Statistics for Machine Learning

Hands-On Exercise Sheet: "Exploring Statistics in the Titanic Dataset"

Objective

Apply key statistical concepts — mean, variance, correlation, hypothesis testing, and regression — to understand survival patterns in the Titanic dataset.

Part 1: Descriptive Statistics

Load the dataset (titanic.csv).

Compute:

1. Mean and median of Age and Fare.
2. Survival rate (Survived column).

Plot a histogram of passenger ages.

```
python
import pandas as pd, matplotlib.pyplot as plt
df = pd.read_csv('titanic.csv')
print(df['Age'].mean(), df['Age'].median())
plt.hist(df['Age'], bins=20, color='skyblue')
plt.title('Age Distribution of Passengers')
plt.show()
```

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Statistics for Machine Learning

Hands-On Exercise Sheet: "Exploring Statistics in the Titanic Dataset"

Part 2: Variance & Correlation

Calculate variance and standard deviation of Fare.

Create a correlation heatmap for selected features (Age, Fare, Sex, Pclass, Survived).

```
python
import seaborn as sns
sns.heatmap(df[['Age', 'Fare', 'Pclass', 'Survived']].corr(), annot=True,
cmap='coolwarm')
```

Part 3: Hypothesis Testing

Test whether women had a higher survival rate than men.

```
python
from scipy.stats import ttest_ind
male = df[df['Sex']=='male']['Survived']
female = df[df['Sex']=='female']['Survived']
print(ttest_ind(female, male))
```

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Interpret the p-value: if $p < 0.05$, the difference is statistically significant.

Statistics for Machine Learning

Hands-On Exercise Sheet: "Exploring Statistics in the Titanic Dataset"

Part 4: Regression

Build a simple logistic regression model to predict survival.

```
python
from sklearn.linear_model import LogisticRegression
from sklearn.preprocessing import LabelEncoder
X = df[['Age', 'Fare', 'Pclass']]
y = df['Survived']
X = X.fillna(X.mean())
model = LogisticRegression()
model.fit(X, y)
print(model.predict([[25, 50, 2]])) # Example passenger
```

Part 5: Reflection

Write a short paragraph (100–150 words) explaining:

- Which statistical measure gave the most insight?
- How did hypothesis testing validate your assumptions?
- How does regression connect statistics to ML prediction?

17

In part 3 answer the last 3 questions

Assignment-4

20/June

To execute the code given here and show the output.

(Code given by official group iict)

Exercise 1: # Linear Regression on Titanic Dataset Python Program code

Exercise 2: Python demo snippet for Logistic Regression on the Titanic dataset, showing how classification differs from regression

Exercise 3: Here's a ready to use Jupyter Notebook template my students can run to experience both Supervised and Unsupervised Learning on the Titanic dataset side by side.

Assignment-5

22/June

LINEAR REGRESSION

Part 3: Scikit-Learn Implementation

EXERCISE FOR 22 JUNE 2026 TO BE DONE

```
Python
from sklearn.linear_model import
LinearRegression

X = X.reshape(-1,1) # reshape for
sklearn
model = LinearRegression()
model.fit(X, y)

print("Slope (m):", model.coef_[0])
print("Intercept (c):",
model.intercept_)
print("Prediction for X=6:",
model.predict([[6]]))
```

**TRY THIS CODE TO LEARN & LET
ME KNOW THE OUTPUT**

Compare results with manual gradient descent.

LINEAR REGRESSION

Part 1: Manual Implementation (Gradient Descent)

Goal: Show how the algorithm learns slope m and intercept c .

```
import numpy as np
import matplotlib.pyplot as plt
```

EXERCISE FOR 22 JUNE 2026 TO BE DONE

```
# Sample dataset
X = np.array([1,2,3,4,5])
y = np.array([2,4,5,4,5])
```

TRY THIS CODE TO LEARN & LET ME KNOW THE OUTPUT

```
# Initialize parameters
m, c = 0, 0
alpha = 0.01
epochs = 1000
```

```
# Gradient Descent
for _ in range(epochs):
    y_pred = m*X + c
    error = y - y_pred
    m -= alpha * (-2 * np.mean(X * error))
    c -= alpha * (-2 * np.mean(error))
```

```
print("Slope (m):", m)
print("Intercept (c):", c)
```

LINEAR REGRESSION

EXERCISE FOR 22 JUNE 2026 TO BE DONE

Part 2: Visualization of Fit

```
python
plt.scatter(X, y, color='blue')
plt.plot(X, m*X + c, color='red')
plt.title("Manual Linear
Regression Fit")
plt.xlabel("X")
plt.ylabel("Y")
plt.show()
```



TRY THIS CODE TO LEARN & LET
ME KNOW THE OUTPUT

Scatter points + regression line = ?

FILL IN THE BLANK & LET KNOW THROUGH CHAT TODAY

Assignment -6

23/June

Answer the following Questions.

Logistic Regression Mini Quiz — Titanic Dataset

Question 1

A 25-year-old passenger in 2nd class has a predicted probability of survival $P = 0.62$.
Classify the outcome (Survived / Did Not Survive).

- Explain why the threshold of 0.5 matters.

Question 2

A 45-year-old passenger in 3rd class has $P = 0.38$.

- What does this probability mean?
- How would the model's decision change if the threshold were 0.4 instead of 0.5?

Question 3

A 10-year-old passenger in 1st class has $P = 0.85$.

- Interpret the probability.
- Discuss how age and class might influence the coefficients β_1 and β_3 .

Bonus Challenge

If gender is added as a variable and the coefficient for "female" is positive, what does that tell us about survival likelihood?

Assignment-7

25/June

DECISION TREES & RANDOM FOREST

Classroom Activity: "Build Your Own Forest"

Part 1 – Decision Tree Challenge

- Objective: Students manually create a mini decision tree.
- Give each group a small dataset (e.g., Titanic passengers or Iris flowers).
 - Ask them to choose one feature to split on (Sex, Age, or Petal Length).
 - They draw branches and predict outcomes.
 - Discuss: *What happens if the tree goes too deep?* → Introduce overfitting.

Part 2 – Random Forest Simulation

- Objective: Experience ensemble voting.
- Each group builds a slightly different tree (using different features).
 - Combine all group predictions for one test case.
 - Majority vote decides the final outcome.
 - Discuss: *How does combining trees improve reliability?*

Part 3 – Reflection

Prompt students:
"If one tree makes a mistake, how does the forest correct it?"
Encourage them to connect this to teamwork — multiple perspectives leading to better decisions.

DECISION TREES & RANDOM FOREST

Worksheet: Data Cleansing with Decision Trees

Section 2 – Tree-Based Cleansing

- Task: Use a Decision Tree to guide corrections.
- Split on Sex, Age, and Fare.
 - Predict missing values or flag outliers.

Feature	Split Condition	Cleansing Action
Sex	male vs Female	Standardize labels
Age	< 10	Impute missing ages
Fare	> 300	Flag as outlier

Reflection: How do tree splits help reveal data issues?

DECISION TREES & RANDOM FOREST

Worksheet: *Data Cleansing with Decision Trees*

Section 1 – Spot the Dirty Data

Task: Identify issues in the sample dataset below.

Passenger	Sex	Age	Fare	Survived
1	male	?	512	0
2	Female	25	71	1
3	M	8	21	1
4	male	35	?	0

Reflection: What kinds of problems do you see? (Missing values, inconsistent entries, outliers)

DECISION TREES & RANDOM FOREST

Classroom Activity: “Build Your Own Forest”

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Part 3 – Reflection

Prompt students:

DECISION TREES & RANDOM FOREST

Worksheet: *Data Cleansing with Decision Trees*

Section 3 – Random Forest Robustness

Task: Discuss how multiple trees handle noisy data.

- Each tree uses different subsets of features.
- Majority vote reduces the effect of errors.

Reflection: Why is ensemble learning more reliable for messy datasets?

Section 4 – Summary

- **Decision Tree:** Detects patterns and anomalies.
- **Random Forest:** Smooths out noise and improves accuracy.
- **Outcome:** Cleaner, more trustworthy data for modeling.

Assignment-8

26/June

UNSUPERVISED LEARNING & NEURAL NETWORK

Student Activity Worksheet: "Exploring the Titanic with AI"

Objective

Understand how unsupervised learning reveals hidden patterns and how neural networks use those insights to predict survival.

Part 1: Data Exploration

Dataset: Titanic.csv (columns: Age, Fare, Sex, Class, Family Size, Survived)

Tasks:

1. Load the dataset in Python or Excel.
2. Remove the *Survived* column — we'll start *unsupervised*.
3. Visualize distributions of Age, Fare, and Class.
4. **Discuss:** *What patterns do you notice?*

Part 2: Clustering

Goal: Group passengers by similarity.

Steps

- Apply K-Means clustering ($k = 3$).
- Label clusters and interpret them:
 - Cluster 1: Wealthy first-class travelers
 - Cluster 2: Families in steerage
 - Cluster 3: Solo young adventurers
- Plot clusters using Age vs. Fare.

Discussion Prompt:

"Which cluster do you think had the highest survival rate?"

UNSUPERVISED LEARNING & NEURAL NETWORK

Student Activity Worksheet: "Exploring the Titanic with AI"

Part 3: Autoencoder Compression

Goal: Reduce data complexity.

Steps:

- Build a simple Autoencoder with 3 layers.
- Encode features into a 2-D latent space.
- Visualize encoded data using a scatter plot.

Discussion Prompt:

"How does compression help us see relationships more clearly?"

Part 4: Feature Discovery with RBM

Goal: Find hidden influences.

Steps

- Train a Restricted Boltzmann Machine on encoded data.
- Extract latent features (e.g., "family influence," "economic status").
- Add these features back to the dataset.

Discussion Prompt:

"What hidden factors might have affected survival?"

UNSUPERVISED LEARNING & NEURAL NETWORK

Student Activity Worksheet: "Exploring the Titanic with AI"

Part 5: Neural Network Prediction

Goal: Predict survival using discovered features.

Steps

- Add the *Survived* column back.
- Train a Feedforward Neural Network.
- Evaluate accuracy and visualize predictions.

Discussion Prompt:

"How did unsupervised learning improve your neural network's performance?"

Reflection

"Unsupervised learning helps us explore the Titanic without knowing who survived. Neural networks learn from those discoveries to predict survival — turning exploration into understanding."

DUE ON 3 July 2026 BY 4 PM TO MY MAIL ID

Assignment-9

27/June

NEURAL NETWORKS, PERCEPTRON & DEEP LEARNING

Worksheet: Understanding Neural Networks and Deep Learning

Part A — Concept Review

1. Define Perceptron in your own words.
2. List three key differences between a Perceptron and a Deep Neural Network.
3. Explain what Gradient Descent does in a neural network.
4. Why is backpropagation essential for learning?

Part D — Think & Reflect

"From simple neurons to deep insights — AI is teaching machines to think."

Write a short paragraph (5–6 lines) on what this statement means to you.

Part B — Titanic Survival Case Study

1. Identify five features used to predict survival in the Titanic dataset.
2. Describe how Gradient Descent helps the model improve its predictions.
3. What patterns might the network learn about survival?
4. Sketch a simple diagram showing how data flows through layers.

Part C — Real-World Applications

Match each domain with its neural-network use case:

Domain	Application
Healthcare	Tumor detection
Finance	Market trend prediction
Space Science	Satellite image analysis
Education	Adaptive learning systems

ASSIGNMENT-10

Date-29/06

Model Evaluation, Accuracy, Precision, Recall, and Confusion Matrix

Applying Accuracy, Precision, Recall & F1-Score to the Titanic Dataset

6. F1-Score

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

- Balances precision and recall.
- Ideal for Titanic because it accounts for both false positives (predicting survival incorrectly) and false negatives (missing actual survivors).

Today’s Assignment

“If you were designing a rescue-prediction system, would you prefer high precision or high recall?”

Model Evaluation, Accuracy, Precision, Recall, and Confusion Matrix

Applying Accuracy, Precision, Recall & F1-Score to the Titanic Dataset

Metric Calculations

Metric	Formula	Interpretation	Titanic Example
Accuracy	$\frac{TP + TN}{TP + TN + FP + FN}$	Overall correctness	$\frac{80+160}{300} = 0.8 \rightarrow 80\%$
Precision	$\frac{TP}{TP + FP}$	How many predicted survivors actually survived	$\frac{80}{100} = 0.8 \rightarrow 80\%$
Recall	$\frac{TP}{TP + FN}$	How many actual survivors were detected	$\frac{80}{120} = 0.67 \rightarrow 67\%$
F1-Score	$2 \times \frac{Precision \times Recall}{Precision + Recall}$	Balance between precision and recall	$2 \times \frac{0.8 \times 0.67}{1.47} = 0.73 \rightarrow 73\%$

Model Evaluation, Accuracy, Precision, Recall, and Confusion Matrix

Classroom Exercise

Provide students with different precision and recall pairs:

Precision	Recall	F1-Score
0.9	0.9	0.9
0.9	0.5	0.64
0.6	0.9	0.72
0.5	0.5	0.5

1. interpret which model is most balanced

Example



If a model has:

- Precision = 0.8
- Recall = 0.6 Then:

$$F1 = 2 \times \frac{0.8 \times 0.6}{0.8 + 0.6} = 0.69$$

So, the model's balanced performance is 69%.

PROJECT TO BE SUBMITTED

PROJECT -1- AI POWERED FAKE NEWS DETECTION USING TEXT CLASSIFICATION

AI-Powered Fake News Detection Using Text Classification

Problem Statement

Build a machine learning pipeline **from scratch** to classify news articles as *real* or *fake*. Students must implement preprocessing, feature extraction, model training, and evaluation without relying on pre-built solutions.

Data Sources

- **Kaggle:** Fake News Detection Dataset ([kaggle.com in Bing](https://www.kaggle.com/datasets/robertblackburn/fake-news-detection))
- **UCI Repository:** Fake News Dataset
- **Optional:** Use *NewsAPI* to scrape recent articles and manually label a subset.

30-Day Workflow

Week

Tasks

- Week 1** Collect dataset, clean text (remove stopwords, punctuation), tokenize manually
- Week 2** Implement Bag-of-Words, TF-IDF, embeddings; exploratory data analysis
- Week 3** Build models (KNN, Logistic Regression, Random Forest, Neural Net)
- Week 4** Evaluate models, visualize results, prepare report & presentation

Algorithms to Implement

- KNN (Non-Parametric)
- Logistic Regression (Parametric)

Algorithms to Implement

I

- **KNN (Non-Parametric)**
- **Logistic Regression (Parametric)**
- **Random Forest (Ensemble)**
- **Simple Neural Network (Deep Learning)**

Python Code Skeleton

```
python
# Week 1: Data Loading & Cleaning
import pandas as pd
import re
from sklearn.model_selection import train_test_split

data = pd.read_csv("train.csv") # Kaggle dataset
X = data['text']
y = data['label']

def clean_text(text):
    text = re.sub(r'\W', ' ', text) # remove punctuation

# Week 2: Feature Engineering
from sklearn.feature_extraction.text import TfidfVectorizer
vectorizer = TfidfVectorizer(max_features=5000)
X_vec = vectorizer.fit_transform(X)

X_train, X_test, y_train, y_test = train_test_split(X_vec, y, test_size=0.2,
                                                    random_state=42)

# Week 3: Model Building
from sklearn.neighbors import KNeighborsClassifier
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier
from sklearn.neural_network import MLPClassifier

models = {
    "KNN": KNeighborsClassifier(n_neighbors=5),
    "LogReg": LogisticRegression(max_iter=1000),
    "RandomForest": RandomForestClassifier(n_estimators=100),
    "NeuralNet": MLPClassifier(hidden_layer_sizes=(100,), max_iter=300)
}

# Train & Evaluate
from sklearn.metrics import accuracy_score, classification_report

for name, model in models.items():
    model.fit(X_train, y_train)
    preds = model.predict(X_test)
    print(f"{name} Accuracy:", accuracy_score(y_test, preds))
    print(classification_report(y_test, preds))
```

I

How AI & ML can be Used with proper expansion of the following

- **AI:** Automates the detection of misinformation.
- **ML:** Learns patterns in text (word frequency, context) to classify articles.
- **NLP:** Preprocessing (tokenization, TF-IDF) converts raw text into numerical features.
- **Evaluation:** Accuracy, precision, recall, F1-score, confusion matrix.

Documentation Guidelines

Documentation Guidelines

Students must submit:

1. **Introduction** — Problem statement, importance of fake news detection.
2. **Dataset Description** — Source, size, features.
3. **Methodology** — Preprocessing, feature extraction, algorithms.
4. **Results** — Accuracy, precision, recall, F1-score, confusion matrices.

5. **Discussion** — Compare parametric vs. non-parametric models.
6. **Conclusion** — Insights, limitations, future scope.
7. **Appendix** — Python code, test data samples.
8. **Format & Standard:** IEEE

PROJECT-2- AI DRIVEN PHISHING EMAIL DETECTION USING NLP(NATURAL LANGUAGE PROCESSING)

Project Title: AI-Driven Phishing Email Detection Using NLP (Natural Language Processing)

Objective

To design and implement a machine-learning system that automatically detects phishing emails using text analysis and metadata features. Students will build the project from scratch, applying data preprocessing, feature extraction, model training, and evaluation.

Scope

- **Collect and preprocess email datasets (Kaggle, UCI, or custom samples).**
- **Extract linguistic and structural features (URLs, sender domains, word patterns).**
- **Train and compare models: Logistic Regression, Random Forest, Naive Bayes, and simple Neural Network.**
- **Evaluate performance using accuracy, precision, recall, and F1-score.**
- **Visualize confusion matrices and feature importance.**

Methodology

Methodology

Phase	Description
Data Collection	Gather phishing and legitimate email samples.
Data Cleaning	Remove HTML tags, punctuation, and stopwords.
Feature Engineering	Apply TF-IDF, word embeddings, and metadata extraction.
Model Development	Train multiple classifiers and tune hyperparameters.
Evaluation	Compare models and interpret results.
Deployment (Optional)	Create a simple web interface for live email classification.